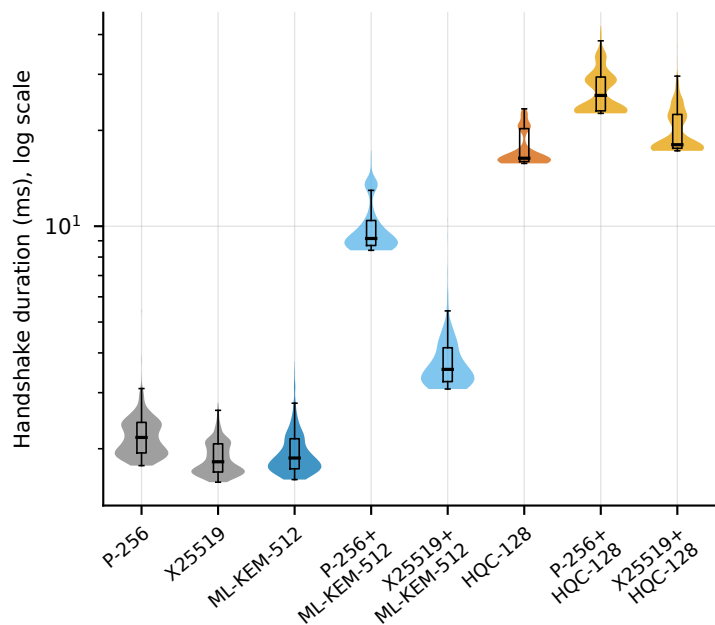


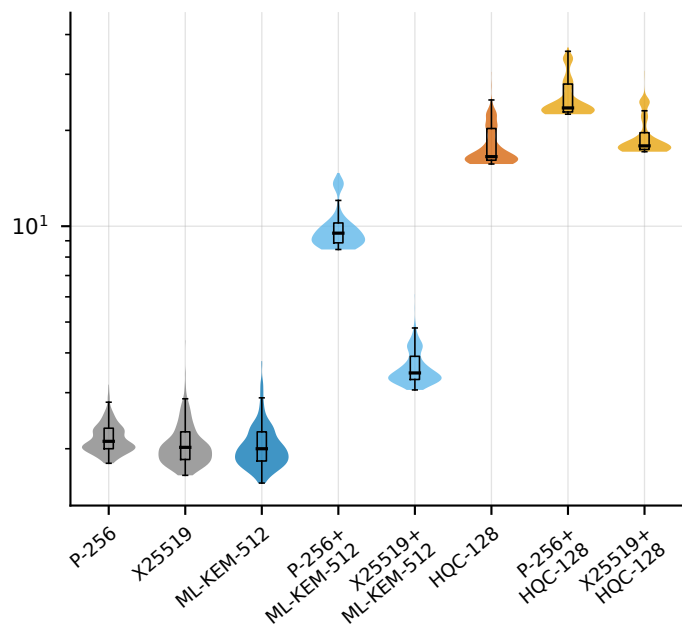
(a)

ed25519 (classical)



(b)

mlDSA44 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlDSA44 Mean (ms)	mlDSA44 CV (%)	mlDSA44 Max (ms)
P-256	2.22	15.2	5.47	2.18	11.0	3.18
X25519	1.89	12.6	2.79	2.10	15.8	4.38
ML-KEM-512	1.97	16.7	3.76	2.09	16.9	3.77
P-256+ML-KEM-512	10.0	19.0	17.3	9.97	15.8	14.7
X25519+ML-KEM-512	3.79	20.2	10.6	3.63	12.5	6.11
HQC-128	17.6	13.0	23.4	18.0	14.4	30.5
P-256+HQC-128	27.0	16.2	42.5	25.5	15.0	36.5
X25519+HQC-128	20.2	16.5	37.5	19.2	13.8	30.8